LOYOLA ACADEMY
A DEGREE & PG COLLEGE
(AUTONOMOUS)
ALWAL, SECUNDERABAD-500010

ACCREDITED BY NAAC WITH ‘A’ GRADE

A COLLEGE WITH POTENTIAL AND EXCELLENCE

B.Sc. COMPUTER SCIENCE AND CYBER SECURITY

THIS IS TO CERTIFY THAT THIS IS A BONAFIDE RECORD ON
NETWORK SECURITY
DONE DURING THE SECOND YEAR, FOURTH SEMESTER FOR THE ACADEMIC YEAR 2025-2026

Name :

Class : NCSCS

UID :

Internal Examiner **Principal** **External Examiner**

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1) **Configure OSPF MD5 Authentication**

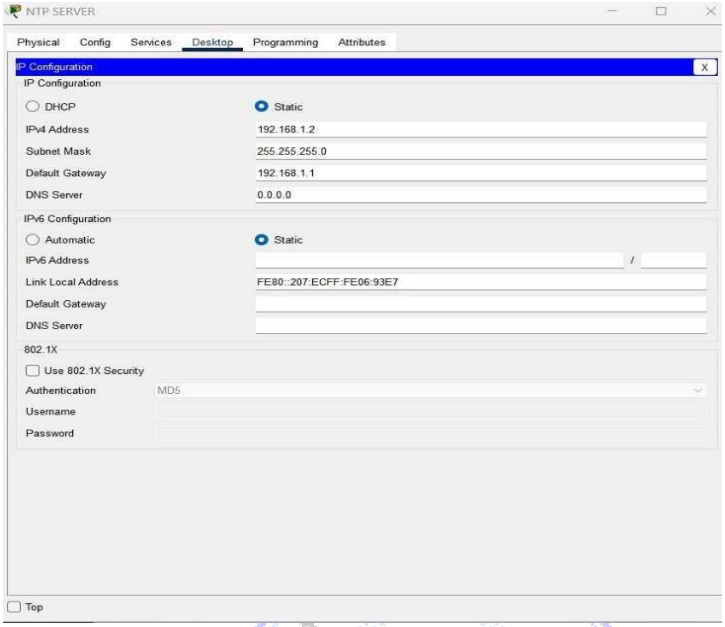
- Open the cisco packet tracer and place 3 routers and rename the display name as R1, R2, R3 and the hostname as the same R1, R2, R3 (In the config tab).
- Then go to R1 – physical tab- switch off the router- place the module name HWIC-2T in the port- switch the router.
- Repeat the same for all the 3 routers.
- Then we need to have the connections between the routers- we need to have serial DTE connection.

  have the connection from R1-R2 (use serial0/1/0) and for R2-R3(use serial0/1/1)
- Then have SWITCHES, SERVERS (SYSLOG SERVER & NTP SERVER) and a PC (SSH CLIENT) in the network and use the auto connectors for connecting the devices.

  Build the network as shown in the figure.

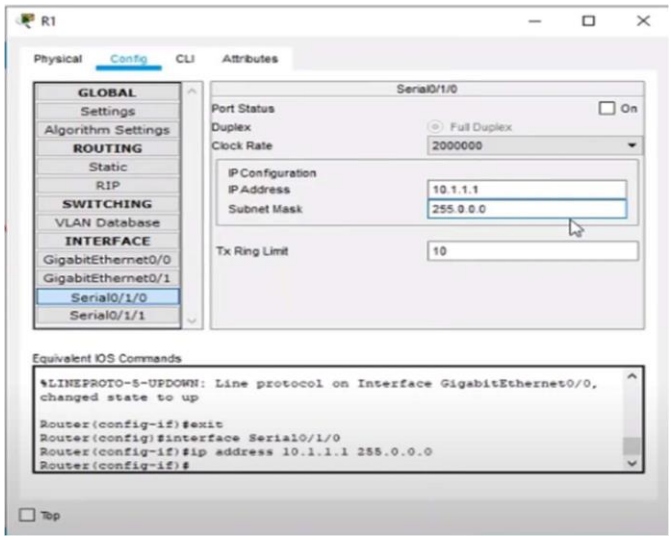
- Now configure the network-
  - Let us have 192.168.1.0 for the first area.
  - Let us have 10.1.1.0 for the second area.
  - Let us have 192.168.2.0 for the third area.
- Check the interface at router 1 to switch0 and check the port gig0/0 and configure the Ip address as 192.168.1.1 and subnet mask as 255.255.255.0 and click on.
- Go to the NTP server and go to the desktop and set the ip address as 192.168.1.2 and subnet mask as 255.255.255.0 and the default gateway will be 192.168.1.1.

- Same for SYSLOG server, go to the desktop- set the Ip address as 192.168.1.3 and the subnet mask as 255.255.255.0 and the default gateway as 192.168.1.1.

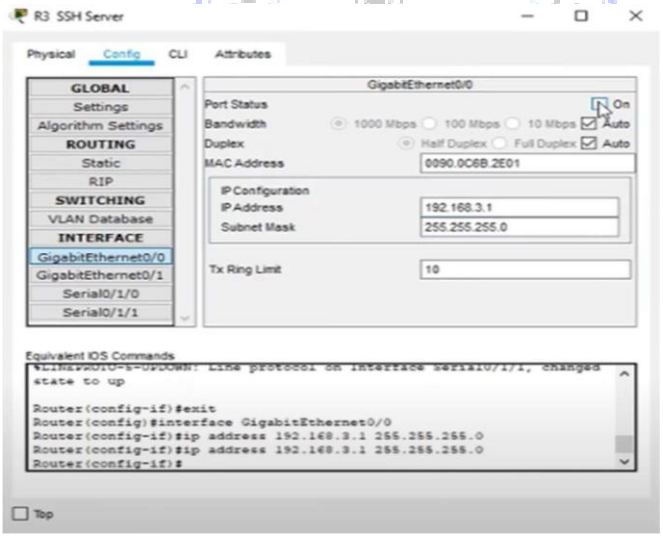


- Now configure routers

  - R1- Go to serial0/1/0 and add the Ip address as 10.1.1.1 and subnet mask as 255.255.255.0 and switch it on.
  - R2- Go to the serial0/1/0 and the Ip address as 10.1.1.2 and the subnet mask as 255.255.255.0 and switch it on.
  - R2- Go to the serial0/1/1 and the Ip address as 10.2.2.1 and subnet mask as 255.255.255.0 and switch it on.
  - R3- Go to the serial0/1/1 and the Ip address as 10.2.2.2 and subnet mask as 255.255.255.0 and switch it on.



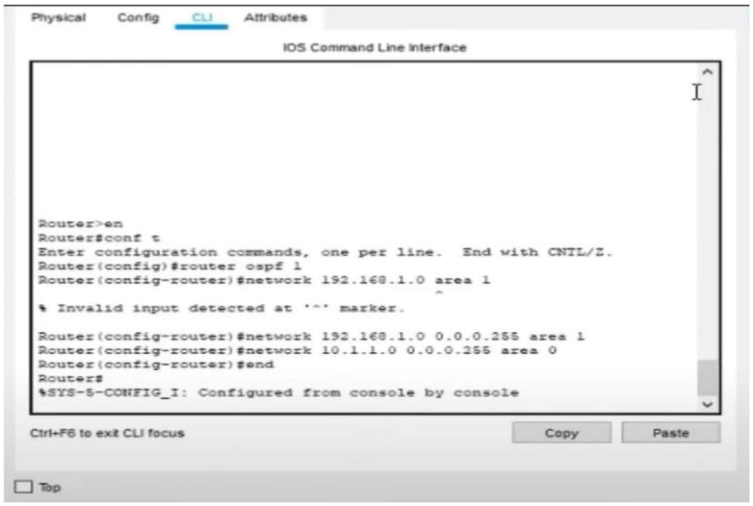
- Now configure the R3-pc area
  - Go to the R3 router and click on connection gig0/1 add the Ip address as 192.168.2.1 and the subnet mask as 255.255.255.0 and switch it ON.
  - The go to PC click on the desktop and add the Ip address as 192.168.2.2 and the subnet mask as 255.255.255.0 and the default gateway as 192.168.2.1.



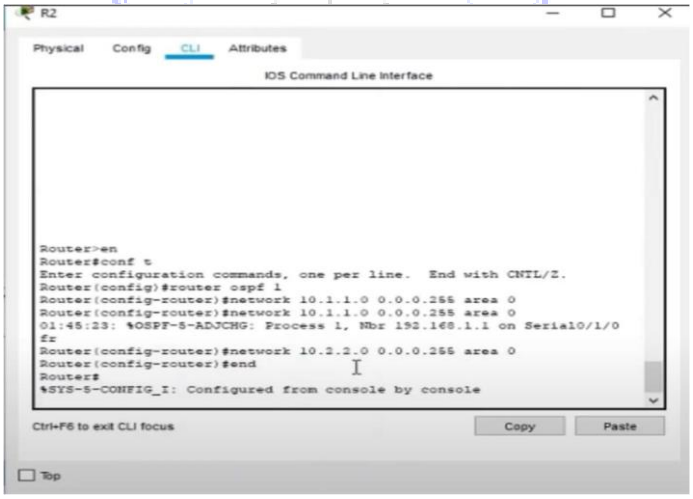
CONFIGURING OSPF ROUTING-

- Go to R1 - Go to CLI tab
  - Router>en
  - Router#conf t
  - Router(config)#router ospf 1
  - Router(config-router)#network 192.168.1.0 0 .0.0.255 area 1
  - Router(config-router)#network 10.1.1.0 0.0.0.255 area 0

- Router(config-router)#end
- Router#

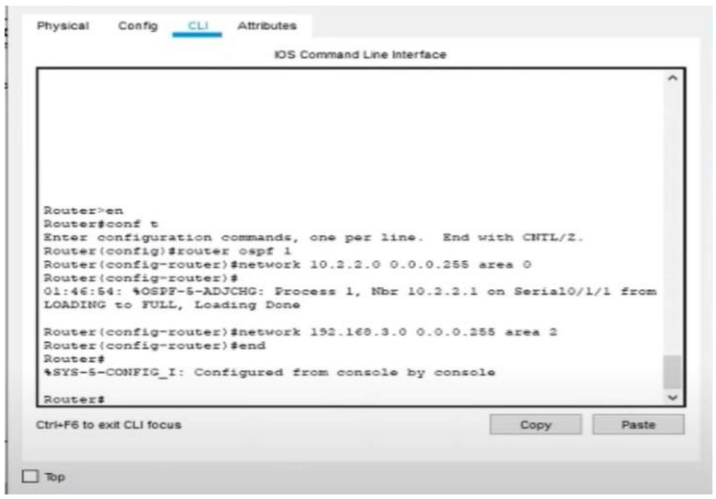


- Go to R2 - Go to CLI tab
  - Router>en
  - Router#conf t
  - Router(config)#router ospf 1
  - Router(config-router)#network 10.1.1.0 0 .0.0.255 area 0
  - Router(config-router)#network 10.1.1.0 0.0.0.255 area 0
  - Router(config-router)#network 10.2.2.0 0.0.0.255 area 0
  - Router(config-router)#end
  - Router#



- Go to R3 - Go to CLI tab
  - Router>en
  - Router#conf t
  - Router(config)#router ospf 1
  - Router(config-router)#network 10.2.2.0 0 .0.0.255 area 0

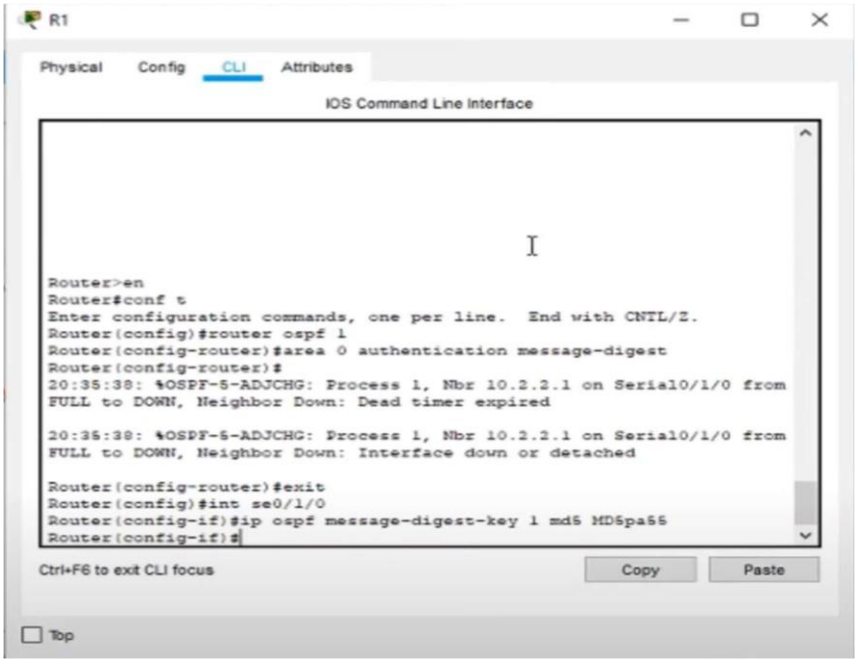
- Router(config-router)#network 192.168.2.0 0.0.0.255 area 2
- Router(config-router)#end
- Router#



CONFIGURE OSPF MD5 AUTHENTICATION.

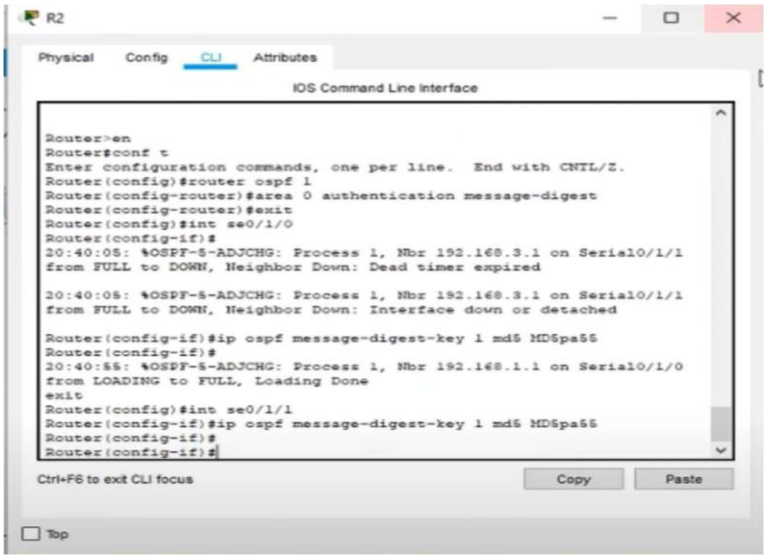
R1- Go to the CLI tab

- Router>en
- Router#conf t
- Router(config)#router ospf 1
- Router(config-router)# area 0 authentication message-digest
- Router(config-router)# exit
- Router(config)# int se0/1/0
- Router(config-if)# ip ospf message-digest-key 1 md5 MD5pa55
- Router(config-if)#



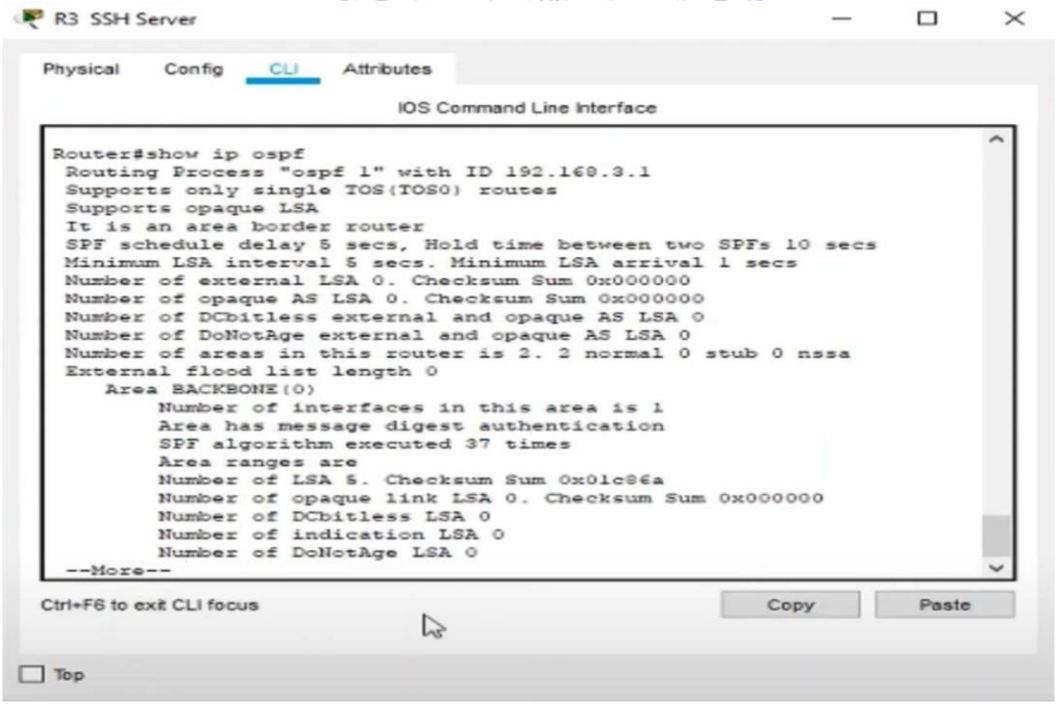
R1- Go to the CLI tab

- Router>en
- Router#conf t
- Router(config)#router ospf 1
- Router(config-router)# area 0 authentication message-digest
- Router(config-router)# exit
- Router(config)# int se0/1/0
- Router(config-if)# ip ospf message-digest-key 1 md5 MD5pa55
- Router(config-if)#exit
- Router(config)# int se0/1/1
- Router(config-if)# ip ospf message-digest-key 1 md5 MD5pa55



R3- Go to the CLI tab

- Router>en
- Router#conf t
- Router(config)#router ospf 1
- Router(config-router)# area 0 authentication message-digest
- Router(config-router)# exit
- Router(config)# int se0/1/1
- Router(config-if)# ip ospf message-digest-key 1 md5 MD5pa55
- Router(config-if)#exit
- Router(config)#exit
- Router# show ip ospf



2) Configure NTP server

- Open the cisco packet tracer and open the same network which we have used before.
- Now enable the NTP server, so click on NTP server and go to services tab- go to NTP and click on enable.
- Enter the key-1 and the password – NTPpa55 and switch it on.
- Go to router R1 and go to CLI tab-
  - Router>en
  - Router#conf t
  - Router(config)#ntp server 192.168.1.2
  - Router(config)# exit
  - Router# show ntp status

```
Router#show ntp status
Clock is unsynchronized, stratum 16, no reference clock
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is
2**24
reference time is 00000000.00000000 (00:00:00.000 UTC Mon Jan 1 1990)
clock offset is 0.00 msec, root delay is 0.00  msec
root dispersion is 0.00 msec, peer dispersion is 0.00 msec.
loopfilter state is 'FSET' (Drift set from file), drift is -
0.000001193 s/s system poll interval is 4, never updated.
Router#
```

  - Router# conf t
  - Router(config)#ntp update-calendar
  - Router(config)#ntp authenticate
  - Router(config)#ntp trusted-key 1
  - Router(config)#ntp authentication-key 1 md5 NTPpa55
  - Router(config)#exit
  - Router#show ntp status

```
Router#
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ntp update-calendar
Router(config)#ntp authenticate
Router(config)#ntp trusted-key 1
Router(config)#ntp authentication-key 1 md5 NTPpa55
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ntp status
Clock is synchronized, stratum 16, reference is 192.168.1.2
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is
2**24
reference time is 1C3AE704.00000398 (0:12:52.920 UTC Sat Mar 18 2051)
clock offset is 0.00 msec, root delay is 0.00  msec
root dispersion is 483.67 msec, peer dispersion is 0.12 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -
0.000001193 s/s system poll interval is 4, last update was 1 sec ago.
Router#
```

- Repeat the same for all the 3 routers.

3) Configure Routers to Log Messages to the Syslog Server

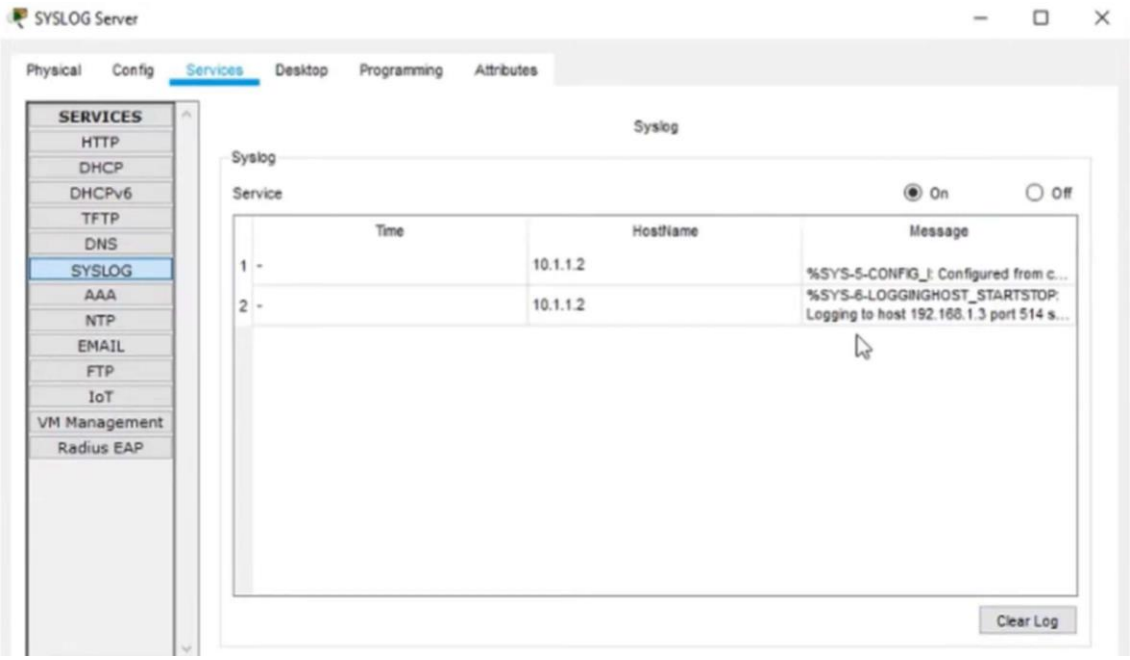
- Open the cisco packet tracer and open the same network which we have used before.
- Click on R1 and go to CLI-
  - Router# conf t
  - Router(config)# logging host 192.168.1.3
  - Router(config)# exit
  - Router# show logging

```
Router#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
          0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.


No Inactive Message Discriminator.


    Console logging: level debugging, 17 messages logged, xml
disabled,
          filtering disabled
    Monitor logging: level debugging, 17 messages logged, xml
disabled,
          filtering disabled
    Buffer logging:  disabled, xml disabled,
          filtering disabled

    Logging Exception size (4096 bytes)
    Count and timestamp logging messages: disabled
    Persistent logging: disabled
```

- Repeat the same for R2
  - Router>en
  - Router# conf t
  - Router(config)# logging host 192.168.1.3
  - Router(config)# exit
  - Router# show logging

```
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
          0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.


No Inactive Message Discriminator.


    Console logging: level debugging, 23 messages logged, xml
disabled,
          filtering disabled
    Monitor logging: level debugging, 23 messages logged, xml
disabled,
          filtering disabled
    Buffer logging:  disabled, xml disabled,
          filtering disabled

    Logging Exception size (4096 bytes)
    Count and timestamp logging messages: disabled
    Persistent logging: disabled

No active filter modules.

 --More--
```

- Now go to the SYSLOG server and click on services- go to SYSLOG and we can see the message are generating.



4) **Configure Router to Support SSH Connections**

- Open the cisco packet tracer and open the same network which we have used before.
- Click on R3 and go to CLI-
  - Router>en
  - Router#conf
  - Router(config)#ip domain-name ccnasecurity.com
  - Router(config)#username SSHadmin privilege 15 secret ciscopa55
  - Router(config)#line vty 0 4
  - Router(config-line)#login local
  - Router(config-line)#transport input ssh
  - Router(config-line)#crypto key generate rsa

Change the router name as R3

  - R3(config)#crypto key generate rsa

How many bits in the modulus [512]:1024

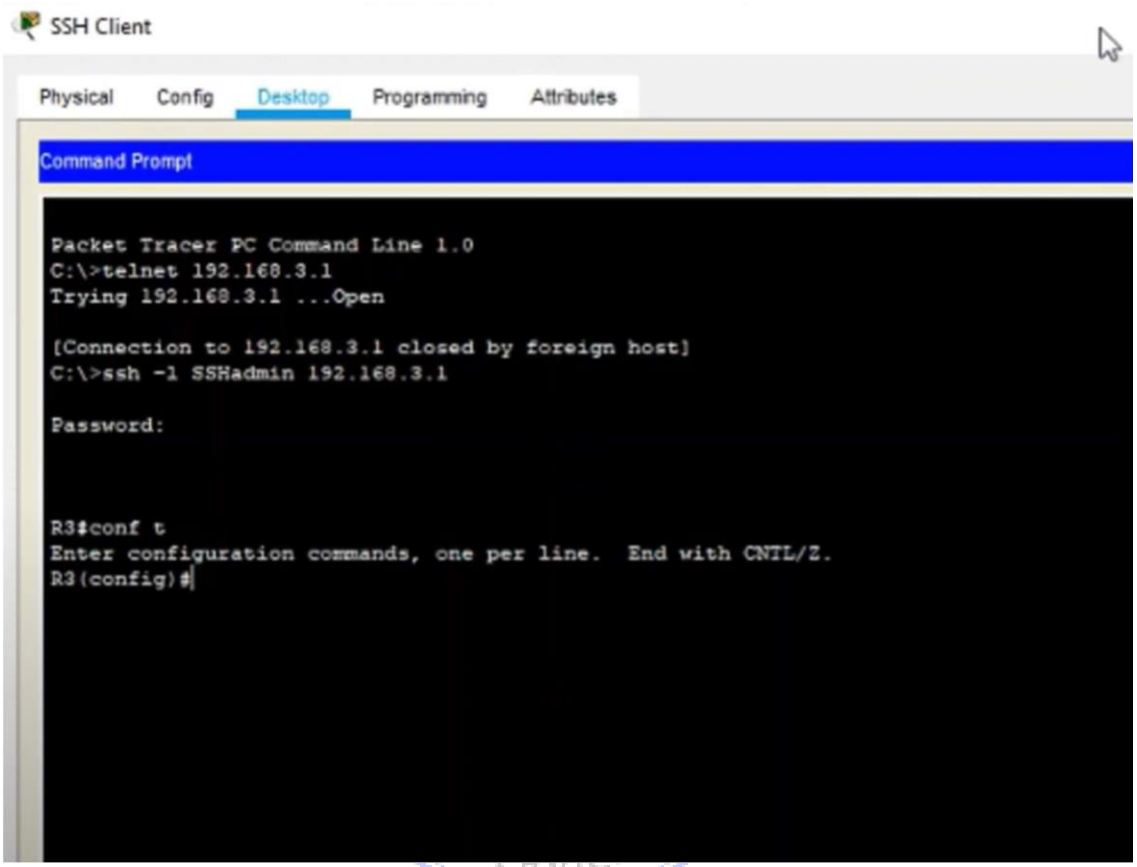
  - R3(config)#exit
  - R3# show ip ssh

```
Router(config)#crypto key generate rsa
% Please define a hostname other than Router.
Router(config)#
Router(config)#hostname R3
R3(config)#crypto key generate rsa
The name for the keys will be: R3.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 2048 for
your
  General Purpose Keys. Choosing a key modulus greater than 512 may
take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R3(config)#exit
*Mar 2 21:4:53.671: %SSH-5-ENABLED: SSH 1.99 has been enabled
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
R3#
```

- Now go to the PC and go to desktop- command prompt
- **C:\telnet 192.168.3.1**

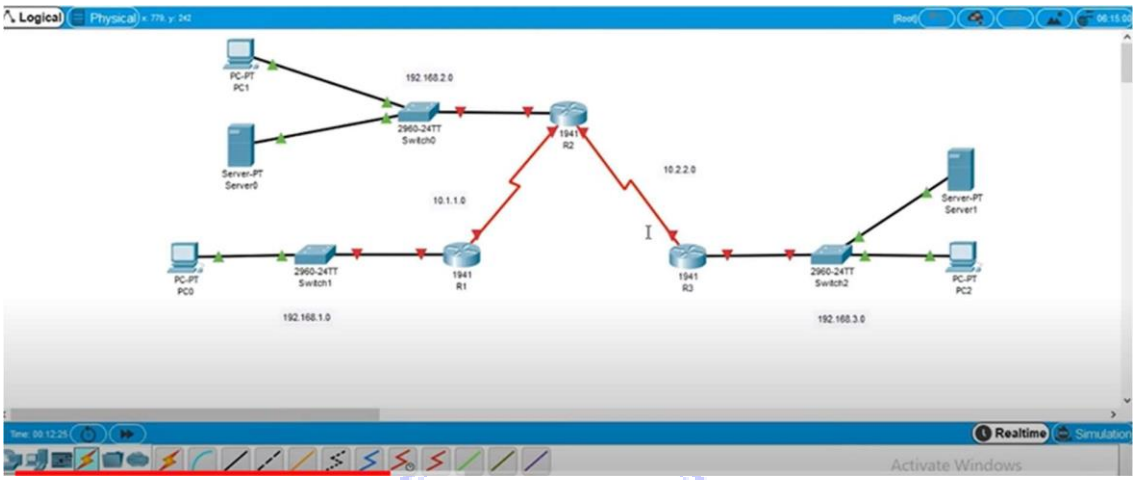
```
Packet Tracer PC Command Line 1.0
C:\>telnet 192.168.3.1
Trying 192.168.3.1 ...Open

[Connection to 192.168.3.1 closed by foreign host]
```

- **C:\>ssh -l SSHadmin 192.168.3.1**
  Password: ciscopa55



5) **Configure server-based AAA authentication**

- Open cisco packet tracer and place 3 routers (1941), 3 switches (2960) , 3 PC and 2 servers and connect the devices as shown in the figure.
- And have the serial connections between routers so add the port (HWIC-2T) to the routers and connect them with serial DTE connection.



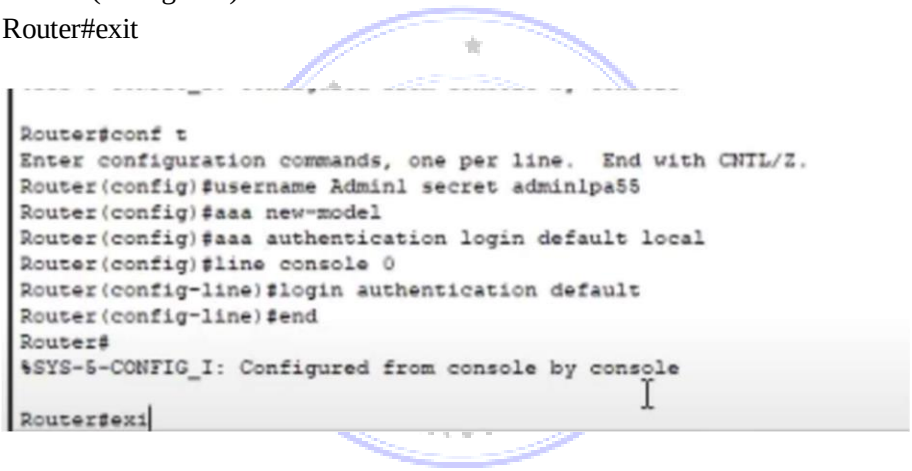
- Go to the router R1 and click on gig0/0 – ip address- 192.168.1.1 & subnet mask- 255.255.255.0 and switch it on.
- Go to pc0- desktop- ip address- 192.168.1.2 & subnet mask- 255.255.255.0 and default gateway as 192.168.1.1
- Now go to R2 and click on gig0/0- ip address-192.168.2.1 and subnet mask is 255.255.255.0 and switch it on.
- Go to pc1- desktop- op address- 192.168.2.2 & subnet mask as 255.255.255.0 and default gateway as 192.168.2.1
- Go to the server- desktop- ip address- 192.168.2.3 & subnet mask as 255.255.255.0 and default gateway as 192.168.2.1.
- Now go to R3 and set the ip configuration in gig0/0- ip address-192.168.3.1 and the subnet mask as 255.255.255.0 and switch it on.
- Now go to PC3 and set the ip config as 192.168.3.2 and subnet mask as 255.255.255.0 and default gateway will be 192.168.3.1
- Go to server- desktop- IP address-192.168.3.3 and the subnet mask as 255.255.255.0 and the default gateway as 192.168.3.1
- Now configure the routers:
    - **(R1- R2)-** serial0/1/0- ip address-10.1.1.1 (R1) & subnet mask- 255.255.255.0(R1)
    - **(R1-R2)-** serial0/1/0-ip address- 10.1.1.2(R2) & subnet mask- 255.255.255.0 (R2)
    - **(R2-R3)-** serial0/1/1- ip address- 10.2.2.1 (R2) & subnet mask- 255.255.255.0 (R2).

- **(R2-R3)-** serial0/1/1- ip address- 10.2.2.2 (R3) & subnet mask- 255.255.255.0 (R3).

- Add RIP to all the routers.
  - **R1-** 192.168.1.0 & 10.1.1.0
  - **R2-** 192.168.2.0, 10.1.1.0 & 10.2.2.0
  - **R3-** 192.168.3.0 & 10.2.2.0

Configure local AAA authentication for console access

- Go to the router R1 – Go to the CLI
  - Router#conf t
  - Router(config)#username Admin1 secret admin1pa55
  - Router(config)#aaa new-model
  - Router(config)#aaa authentication login default local
  - Router(config)#line console 0
  - Router (config-line)# login authentication default
  - Router (config-line)# end
  - Router#exit

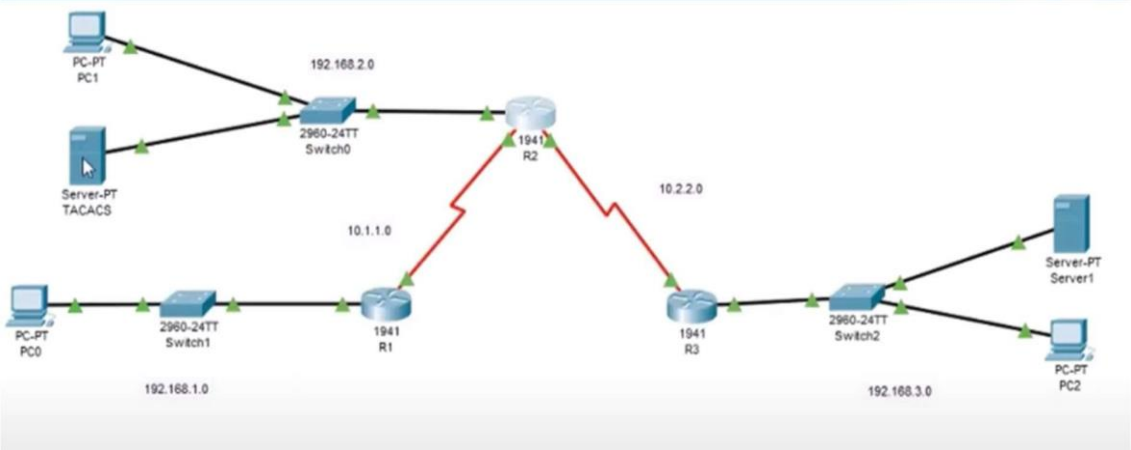


USERNAME: Admin1

PASSWORD: admin1pa55

6) Configure server-based AAA authentication using TACACS+

- Go to the same network of AAA which we have configured before and rename the server as TACACS.



- Click on router R2 and go to CLI tab
  - R2#conf t
  - R2(config)#username Admin2 secret admin2pa55
- Now go to the TACACS sever and go to services tab- click on AAA
  Client-name – R2
  Client ip- 192.168.2.1
  Secret – 654321
  Servertype- tacacs
  Now click on add

- User setup-
  Username- Admin2
  Password- admin2pa55
  Now switch on the server.

- Go to R2- CLI mode
  - R2(config)#ip hostname R2
  - R2(config)#tacacs-server host 192.168.2.3
  - R2(config)#tacacs-server key 654321
  - R2(config)#aaa new-model
  - R2(config)#aaa authentication login default group tacacs+ local
  - R2(config)#line console 0
  - R2(config-line)#login authentication default
  - R2(config-line)#end
  - R2(config)#exit

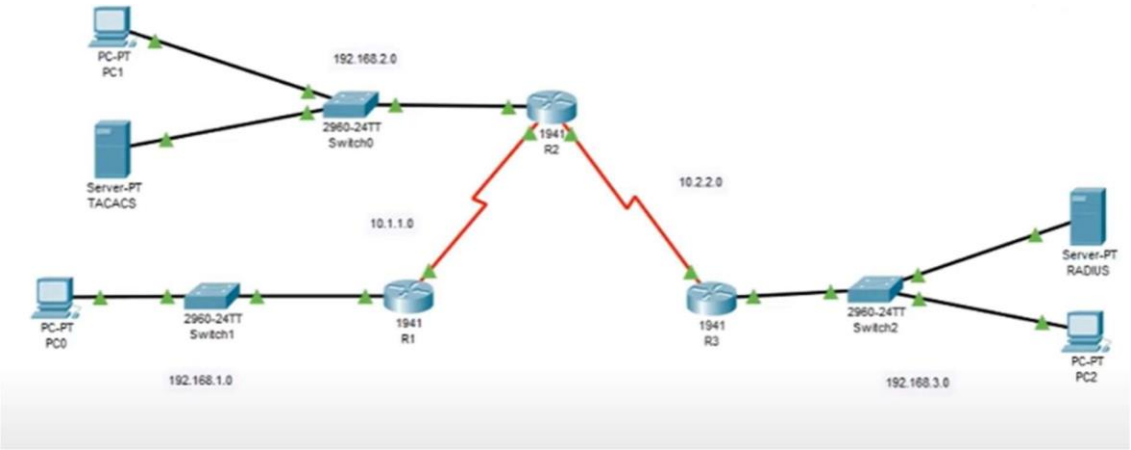
```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#username Admin2 secret admin2pa55
R2(config)#tacacs-server host 192.168.2.3
R2(config)#tacacs-server key 654321
R2(config)#aaa new-model
R2(config)#aaa authentication login default group tacacs+ local
R2(config)#line console 0
R2(config-line)#login authentication default
R2(config-line)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

USERNAME: Admin2

PASSWORD: admin2pa55

7) Configuring Server-Based AAA Authentication Using RADIUS

- Go to the same network of AAA which we have configured before and rename the server as RADI US.



- Go to R3 router- CLI tab
  - R3>
  - R3>en
  - R3# conf t
  - R3(config)#username Admin3 secret admin3pa55
- Go to RADIUS server- go to services – click on AAA
- Network configuration
  Client name- R3
  Client ip- 192.168.3.1
  Secret- 123456
  Servertype- radius
  Now click on add
- User setup
  Username-Admin3
  Password- admin3pa55
  And click on add and switch on the server
- Go to R3 router- CLI tab
  - R3(config)#radius-server host 192.168.3.3
  - R3(config)#raduis-server key 123456
  - R3(config)#aaa new-model
  - R3(config)#aaa authentication login default group radius local
  - R3(config)#line console 0
  - R3(config-line)#login authentication default
  - R3(config-line)#end

- R3#exit

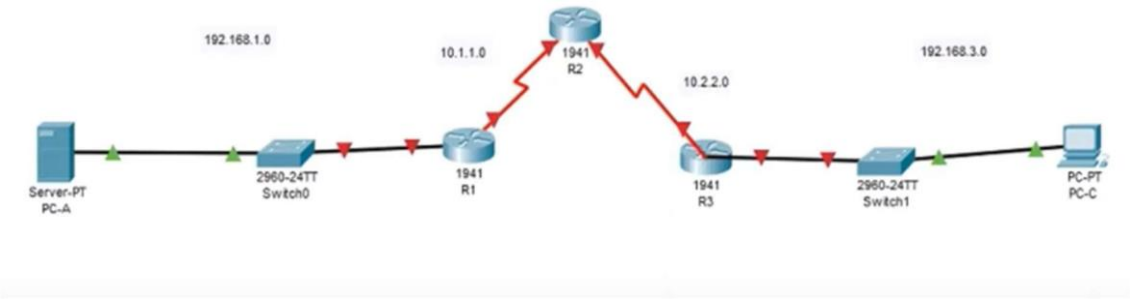
```
R3>
R3>en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#username Admin3 secret admin3pa55
R3(config)#radius-server host 192.168.3.3
R3(config)#radius-server key 123456
R3(config)#aaa new-model
R3(config)#aaa authentication login default group radius local
R3(config)#line console 0
R3(config-line)#login authentication default
R3(config-line)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console
```

USERNAME: Admin3

PASSWORD: admin3pa55

8) **Configuring ACLs**

- Open cisco packet tracer and place 3 routers (1941), 2 switches (2960) , 1 PC and 1 servers and connect the devices as shown in the figure
- And have the serial connections between routers so add the port (HWIC-2T) to the routers and connect them with serial DTE connection.



- Go to the router R1 and click on gig0/0 – ip address- 192.168.1.1 & subnet mask- 255.255.255.0 and switch it on.
- Go to server (PC-A)- desktop- ip address- 192.168.1.2 & subnet mask- 255.255.255.0 and default gateway as 192.168.1.1
- Now go to R3 and click on gig0/0- ip address-192.168.3.1 and subnet mask is 255.255.255.0 and switch it on.
- Go to pc-C- desktop- op address- 192.168.3.2 & subnet mask as 255.255.255.0 and default gateway as 192.168.3.1
- Now configure the routers:
  - **(R1- R2)-** serial0/1/0- ip address-10.1.1.1 (R1) & subnet mask- 255.255.255.0(R1)
  - **(R1-R2)-** serial0/1/0-ip address- 10.1.1.2(R2) & subnet mask- 255.255.255.0 (R2)
  - **(R2-R3)-** serial0/1/1- ip address- 10.2.2.1 (R2) & subnet mask- 255.255.255.0 (R2).
  - **(R2-R3)-** serial0/1/1- ip address- 10.2.2.2 (R3) & subnet mask- 255.255.255.0 (R3).
- Now set the loopback Ip address- Go to the cli
  - Router(config)#interface loopback 1
  - Router(config-if)#ip address 192.168.2.1 255.255.255.0
  - Router(config)#exit
- Add RIP to all the routers.
  - **R1-** 192.168.1.0 & 10.1.1.0
  - **R2-** 192.168.2.0, 10.1.1.0 & 10.2.2.0
  - **R3-** 192.168.3.0 & 10.2.2.0

```
Router(config-if)#
%LINK-5-CHANGED: Interface Loopback1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed
state to up

Router(config-if)#ip address 192.168.2.1
% Incomplete command.
Router(config-if)#ip address 192.168.2.1 ?
  A.B.C.D  IP subnet mask
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#exit
Router(config)#
```

- Go to R1 – CLI

  R1(config)#ip hostname R1

  - R1(config)#ip domain-name ccnasecurity.com
  - R1(config)#username Admin1 privilege 15 secret cisco1
  - R1(config)#line vty 0 4
  - R1(config-line)#login local
  - R1(config-line)#transport input ssh
  - R1(config-line)#exit
  - R1(config)#crypto key generate rsa

  how many bits in the module [512] : 1024

  - R1(config)#exit
  - R1# show ip ssh

```
R1(config)#username Admin1 privilege 15 secret cisco1
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#transport input ssh
R1(config-line)#exit
R1(config)#crypto key generate rsa
The name for the keys will be: R1.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 2048 for
your
  General Purpose Keys. Choosing a key modulus greater than 512 may
take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#exit
*Mar 1 0:15:20.222: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
R1#
```